

Bindita Chaudhuri

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Summary

Senior Applied Research Scientist with 10+ years of expertise (4+ years industrial R&D) steering the development and production deployment of multimodal Generative AI systems (Audio, Video, 3D Humans, Neural Rendering).

Proven track record of leading technical roadmaps and scaling cutting-edge models into commercial products at Meta, Microsoft, and Flawless AI. Ph.D. in Computer Vision with deep technical expertise in DiTs, Gaussian Splatting, and NeRFs, backed by primary-author publications at top-tier venues (CVPR, ECCV, ICCV).

Expertise includes multimodal foundation models for audio-visual synthesis, scene text editing using DiTs, speech-driven facial animation, neural avatars using GANs, controllable human motion generation and photorealistic 3D reconstruction.

Technical Skills

- **Deep Learning:** PyTorch, TensorFlow, Keras.
- **Generative AI:** Diffusion Transformers (large-scale distributed training), GANs, VAEs, NeRFs, 3D Gaussian Splatting.
- **Core Languages & Tools:** Python, C/C++, CUDA, Swift.
- **Domains:** Multimodal (Audio-Visual) Foundation Models, 3D Computer Vision, Digital Humans, Facial Animation.

Work Experience

Flawless AI, Senior Applied Scientist/Applied Science Manager

2023 - Present
Los Angeles, CA

- Developed **multilingual audio-driven 3D facial animation** pipeline with expressive speaking styles using **audio-visual foundation models**.
- **Improved photorealism and temporal consistency** of the neural rendering GAN pipeline for Hollywood-grade filmmaking and visual effects. Also improved **3D face modeling using** high-dimensional **Gaussian Splatting**.
- Hardened the neural rendering pipeline against edge cases (occlusions, facial hair, extreme poses), improving production-grade robustness by $[X]\%$.
- Designed **text-controlled 3D human body motion generation and editing** frameworks, lowering manual animation time for artists (*ICCV 2025*).
- Architected an end-to-end **video scene text editing** pipeline using OCR, SAM2 and DiTs, establishing a new product capability for automated localization.
- **Led a team of 5+ scientists and interns**, defining the technical roadmap from foundational research to production deployment.
- **Coordinated priorities across engineering and product teams** to setup a real-time inference and distributed finetuning pipeline for neural rendering in artist-facing tools.

Meta, Research Scientist

2021 – 2023
Redmond, WA

- Scaled and optimized large-scale Generative Transformers (DiTs) for text/image-to-video synthesis pipelines, improving distributed training efficiency / throughput by $[X]\%$.
- Designed TalkinNeRF, an animatable NeRF for reconstructing **speech-conditioned full-body humans** from text and speech inputs (*ECCVW 2024*).
- Developed a novel multimodal **coarse-to-fine approach** for **detailed 4D face geometry reconstruction** from in-the-wild videos for AR/VR applications (*published in CVPR 2023*).
- Created a **spatio-temporal deep network** to accumulate information from dynamic in-the-wild videos, achieving self-occlusion-free **face texture completion**.
- **Advanced speech-driven 3D facial animation** by integrating neural processing of emotion and non-speech vocalizations, drastically improving realism.
- **Enhanced texture recovery** for synthetic faces from challenging videos utilizing a highly customized StyleGAN-based architecture.

Facebook Reality Labs Research, Research InternSummer 2020
Sausalito, CA

- Designed a **region-adaptive VAE** to synthesize photorealistic texture maps for 3D humans with independent geometry and appearance controllability targeting **virtual try-on** applications and semi-supervised **synthetic data** generation (CVPR 2021).

Microsoft Cloud & AI, Student Researcher2019 - 2020
Redmond, WA

- Developed a **high-fidelity personalized 3D face avatar generation** pipeline from user videos, reducing identity drift (ECCV 2020 *Spotlight*).
- Stabilized real-time face tracking** algorithms optimized for challenging, unconstrained in-the-wild video streams.

Microsoft Research, Research InternSummer 2018
Redmond, WA

- Designed a real-time multi-task learning framework for joint 2D face detection and 3D facial motion retargeting (CVPR 2019); **successfully shipped the model to production for [SwiftKey Puppets](#)**, scaling to **millions of active Android users** for immersive virtual communication.

Intel Labs, Research InternSummer 2017
Santa Clara, CA

- Proposed a Spatial Transformer Network for optical flow prediction and image super-resolution to enable high-quality frame interpolation from HD multi-camera arrays. U.S. [patent](#) granted.

Education

University of Washington, Computer Science

2016 – 2021

Ph.D.

- Thesis*: Deep Facial Expression Modeling and 2D-to-3D Motion Retargeting

Indian Institute of Technology Bombay, Electrical Engineering

2014 – 2016

M.E.

- Honors*: Department Academic Excellence Award
- Thesis*: Region-based Remote Sensing Image Retrieval using Graph Theory

Jadavpur University, Electronics and Telecommunication Engineering

2010 – 2014

B.E.

- Honors*: University Gold Medal, The Supriya Basu Scholarship & others
- Thesis*: Low Cost Low Bandwidth Virtual Education Platform Design

Selected Publications

KinMo: Kinematic-aware Human Motion Understanding and Generation [[link](#)]

ICCV 2025

TalkinNeRF: Animatable Neural Fields for Full-Body Talking Humans [[link](#)]

ECCVW 2024

Semi-supervised Synthesis of High-Resolution Editable Textures for 3D Humans [[link](#)]

CVPR 2021

Personalized Face Modeling for Improved Face Reconstruction and Motion Retargeting [[link](#)]ECCV 2020 (**Spotlight**)Joint Face Detection and Facial Motion Retargeting for Multiple Faces [[link](#)]

CVPR 2019

Learning to Generate 3D Stylized Character Expressions from Humans [[link](#)]

WACV 2018

Deep Learning for Multi-Camera View Interpolation [[link](#)]

U.S. Patent 10,547,823, 2020

Professional Activities

- Conference Reviewer**: CVPR, ECCV, ICCV, NIPS, SIGGRAPH Asia, BMVC, ICLR, ICPR, IEEE VR
- Journal Reviewer**: ACM TOG, IEEE TPAMI, CVIU, IEEE TGRS, IEEE GRSL
- Invited Talk**: Beyond PhD Conference, University of Southern California
- Area Chair (Student)**: CSE Graduate Admissions Committee, University of Washington (2020).
- Teaching Assistant**: University of Washington (CSE) and IIT Bombay (EE).
- People's Choice Award (University of Washington Research Day, 2017)